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(SCOPE)

BCSE414L - HIGH PERFORMANCE COMPUTING

SUBMITTED TO : Prof. MANJULA V

DIGITAL ASSIGNMENT - 1

TOPIC: ALZHEIMER'S PREDICTION OPTIMIZATION USING HPC

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ALZHEIMER'S PREDICTION OPTIMIZATION USING HPC

1) ABSTRACT:

"Alzheimer's disease prediction relies on analyzing complex, high-dimensional data, such as medical imaging and genetic information, often hindered by computational inefficiencies. This work proposes an innovative solution leveraging High-Performance Computing (HPC) to optimize machine learning models, enabling faster processing and improved predictive accuracy. By integrating parallel processing, automated hyperparameter tuning, and advanced HPC frameworks, the approach addresses scalability and computational bottlenecks. Unlike traditional methods, this solution reduces execution time, enhances model performance, and supports the analysis of larger datasets. The proposed methodology offers a transformative step forward in early Alzheimer's diagnosis, with the potential to revolutionize diagnostic workflows and facilitate personalized treatment strategies."

2) PROBLEM STATEMENT:

Alzheimer's disease is a progressive neurological disorder that significantly impacts memory, cognition, and daily functioning. Early diagnosis is critical for timely interventions, yet current predictive approaches face limitations due to the complexity and high-dimensional nature of medical imaging data. Existing methods often struggle with scalability, computational inefficiencies, and suboptimal accuracy when applied to large datasets. For instance, the dataset for this study contains over 44,000 MRI images categorized into four classes of disease severity: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented. Processing such a large volume of data poses significant computational challenges, making it difficult to implement timely and accurate predictive models.

This project aims to address these challenges by leveraging High-Performance Computing (HPC) to optimize machine learning models. The integration of parallel processing and automated hyperparameter tuning will help overcome computational bottlenecks, enhance scalability, and improve predictive accuracy. This innovation seeks to transform the early diagnosis and treatment landscape for Alzheimer's disease.

OBJECTIVES:

Data Analysis and Preprocessing

- Analyze the dataset comprising 44,000 MRI images across four classes of disease severity.
- Implement preprocessing techniques to prepare the data for machine learning models, including noise reduction, normalization, and augmentation.

Model Development and Optimization

- Develop machine learning models to classify MRI images based on Alzheimer's severity levels.
- Utilize HPC to integrate parallel processing and automated hyperparameter tuning to improve model efficiency and accuracy.

Scalability and Computational Efficiency

- Leverage HPC frameworks to ensure scalability for processing large datasets.
- Minimize execution time for model training and inference without compromising accuracy.

Performance Evaluation

- Evaluate the models using metrics such as accuracy, precision, recall, F1-score, and computational efficiency.
- Compare the proposed solution against existing approaches to highlight improvements in predictive performance and execution time.

Impact on Early Diagnosis

- Demonstrate the potential of the optimized models to support early Alzheimer's diagnosis, facilitating timely interventions and personalized treatment planning.

3) LITERATURE REVIEW :

(i) a) Title - Adaptive 3DCNN-Based Interpretable Ensemble Model for Early Diagnosis of Alzheimer's Disease (1)
b) Author - Dan Pan, Genqiang Luo, An Zeng, Chao Zou, Haolin Liang, Jianbin Wang, Tong Zhang, and Baoyao Yang
c) Date of publication - 30 November 2022
d) Publisher Information - IEEE Transactions on Computational Social Systems (Volume: 11, Issue: 1)

(ii) Problem Statement

Alzheimer's disease (AD) and mild cognitive impairment (MCI) are progressive neurological disorders that significantly affect memory and cognitive functions. Early and accurate diagnosis is critical for timely intervention, yet traditional diagnostic methods face challenges in identifying subtle patterns in brain imaging data. Furthermore, most machine learning models for AD diagnosis lack interpretability, which is essential for clinical adoption. Although deep learning methods have shown promise, they often fail to identify specific brain regions contributing to their predictions, limiting their usefulness for understanding disease progression. This project aims to address these limitations by developing an adaptive, interpretable ensemble model combining 3D Convolutional Neural Networks (3DCNN), ensemble learning (EL), and genetic algorithms (GA) to enhance diagnostic accuracy and interpretability.

Objectives

1. **Development of Adaptive Ensemble Model:**
    - Create a hybrid 3D CNN+EL+GA model to differentiate between healthy individuals, MCI patients, and those with AD using neuroimaging data.
2. **Behavioral Domain Mapping:**
    - Analyze the relationship between identified brain regions and their corresponding behavioral domains, linking neuroimaging patterns with cognitive functions impaired in AD.
3. **Evaluation of Model Performance:**
    - Benchmark the model's performance against state-of-the-art deep learning algorithms on datasets like ADNI and OASIS.
4. **Generalizability Exploration:**

- Investigate the applicability of the proposed model in identifying discriminative brain regions for other neurological and psychiatric disorders.

iii) Methodology Adopted

1. **Data Collection and Preprocessing:**
    - Use datasets from the AD Neuroimaging Initiative (ADNI) and Open Access Series of Imaging Studies (OASIS).
    - Preprocess MRI images, including normalization, voxel-level segmentation, and augmentation, to prepare data for the 3DCNN model.
2. **Model Development:**
    - Design a 3D Convolutional Neural Network to extract spatial features from MRI images.
    - Integrate ensemble learning to improve robustness and generalization.
    - Utilize genetic algorithms (GA) for feature selection and hyperparameter optimization to adaptively enhance model performance.
3. **Interpretability Framework:**
    - Apply gradient-based attribution techniques to locate discriminative brain subregions at the voxel level.
    - Visualize significant brain regions contributing to the model's predictions.
4. **Behavioral Analysis:**
    - Map identified brain regions to behavioral domains using neuroanatomical knowledge, linking them to cognitive functions impaired in early AD stages.
5. **Performance Evaluation:**
    - Evaluate the model's performance using metrics like accuracy, precision, recall, and F1-score.
    - Compare results with other state-of-the-art deep learning models.

iv) Performance Metrics

Accuracy

- Measures the overall correctness of the model in classifying Alzheimer's stages (e.g., Non-Demented, Very Mild Demented, Mild Demented).

Sensitivity (Recall)

- Determines the model's ability to correctly identify subjects with Alzheimer's Disease (AD) or Mild Cognitive Impairment (MCI).

Specificity

- Evaluates the model's ability to correctly classify non-demented individuals.

F1-Score

- Balances Precision and Recall to evaluate performance, especially with imbalanced datasets like ADNI and OASIS.

Voxel-Level Discrimination Accuracy

- Validates how well the model identifies discriminative brain subregions contributing to classifications, e.g., rostral hippocampus.

AUC-ROC (Area Under the Receiver Operating Characteristic Curve)

- Measures the trade-off between sensitivity and specificity across different thresholds, providing a comprehensive assessment of model performance.

Behavioral Domain Consistency

- Assesses whether identified brain regions correlate with functional impairments (e.g., emotion, cognition) known in AD progression.

Computational Efficiency

- Includes **training time** and **inference time**, highlighting the model's scalability and feasibility for large datasets (e.g., ADNI, OASIS).

Limitations

1. **Data Dependency:**
    - The model's performance heavily relies on the quality and diversity of neuroimaging datasets. Limited data from rare disorders may hinder generalizability.
2. **Computational Requirements:**

    - Training 3DCNN models with genetic algorithms is computationally expensive and may require high-performance computing resources.
3. **Generalizability Across Disorders:**
    - While promising, the method's applicability to other brain disorders needs further validation with diverse datasets and disease-specific nuances.
4. **Clinical Integration:**
    - Interpretability tools, while valuable, may still lack the granularity needed for direct clinical adoption and real-time decision-making.
5. **Voxel-Level Analysis:**
    - The voxel-level resolution increases the model's complexity and may introduce noise, potentially impacting accuracy if not managed properly.

LITERATURE SURVEY:

EXISTING SYSTEMS

1. **Traditional Methods:**
    - Machine learning models with handcrafted features.
    - Low accuracy and limited ability to process large, high-dimensional datasets like MRI scans.
2. **2D CNNs for Alzheimer's Diagnosis:**
    - These models analyze 2D slices of MRI scans but lose 3D spatial information, resulting in suboptimal performance.
3. **Black-Box Neural Networks:**
    - Many state-of-the-art neural networks provide high accuracy but lack interpretability, making them less suitable for clinical use.
4. **Feature Selection Methods:**
    - Many existing models lack robust feature selection mechanisms, leading to noise in classification tasks.

GAPS IDENTIFIED

1. **Inadequate Use of 3D Spatial Information:**
    - Existing approaches fail to fully utilize the spatial relationships in 3D MRI scans.
2. **Lack of Model Interpretability:**

    - Models fail to explain how decisions are made or which brain regions contribute to predictions.
3. **Insufficient Scalability:**
    - Computational inefficiencies limit the application of existing methods to larger datasets or across multiple disorders.
4. **Poor Linkage Between Brain Regions and Behavioral Domains:**
    - Current methods do not effectively connect neuroimaging patterns with cognitive or behavioral impairments caused by AD.
5. **Limited Generalizability:**
    - Models often perform well on specific datasets but fail to generalize to others like ADNI or OASIS.

PROPOSED METHOD

Adaptive 3D CNN-Based Interpretable Ensemble Model (3DCNN+EL+GA)

1. **3D Convolutional Neural Network (3DCNN):**
    - Extracts spatial and voxel-level features from 3D MRI scans, leveraging the full volumetric data.
2. **Ensemble Learning (EL):**
    - Combines multiple 3DCNN models to improve robustness, reduce overfitting, and enhance generalization across datasets.
3. **Genetic Algorithm (GA):**
    - Optimizes hyperparameters and performs feature selection to identify the most relevant features for classification.
4. **Gradient-Based Attribution Method:**
    - Enhances model interpretability by identifying and visualizing discriminative brain regions (e.g., hippocampus, rostral hippocampus) at a voxel level.
5. **Behavioral Domain Mapping:**
    - Links identified brain regions to cognitive and behavioral functions, providing insights into AD-related impairments.
6. **Generalization Across Disorders:**
    - Extends the methodology to other neurological disorders, such as Parkinson's disease and autism, using neuroimaging data.

DATASETS DESCRIPTION

Alzheimer's Disease Neuroimaging Initiative (ADNI):

- **Description:**
  - A publicly available dataset containing neuroimaging and clinical data for Alzheimer's research.
- **Data Structure:**
  - Includes MRI scans labeled as Non-Demented, Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD).
- **Usage:**
  - Used for training and validating the proposed model.

Open Access Series of Imaging Studies (OASIS):

- **Description:**
  - Another freely available neuroimaging dataset designed for studying brain health.
- **Data Structure:**
  - Includes MRI scans labeled as Non-Demented and Demented.
- **Usage:**
  - Used to evaluate the generalizability of the proposed model.

SAMPLE DATA:

Comparison of Experimental Results in OASIS Datasets

Experiment	Method	Accuracy	AUC	Recall	Precision	F1-score
AD vs. NC	2DCNN+EL [14]	0.73±0.01	0.86±0.01	0.62±0.02	0.92±0.03	0.74±0.01
	2DCNN+EL+GA [16]	0.72±0.02	0.86±0.02	0.58±0.02	0.94±0.02	0.72±0.02
	vanilla 3DCNN+EL	0.74±0.03	0.75±0.02	0.58±0.05	0.89±0.01	0.74±0.03
	3DCNN+EL+Algo.1	0.74±0.005	0.52±0.02	0.56±0.01	0.91±0.01	0.69±0.009
	3DCNN+EL+PSO	0.75±0.01	0.74±0.01	0.64±0.04	0.78±0.04	0.74±0.01
	3DCNN+EL+PSO2019	0.77±0.02	0.76±0.02	0.62±0.05	0.84±0.02	0.76±0.02
	3DCNN+EL+GA	0.76±0.02	0.75±0.02	0.61±0.04	0.83±0.02	0.75±0.02

Comparison of Experimental Results in ADNI Datasets

Experiment	Method	Accuracy	AUC	Recall	Precision	F1-score
AD vs. NC	PCA+SVM [20]	0.76±0.11	—	—	—	—
	2DCNN [15]	0.77±0.07	—	—	—	—
	2DCNN+EL [14]	0.84±0.05	0.92±0.03	—	—	—
	2DCNN+EL+GA [16]	0.86±0.05	0.92±0.04	—	—	—
	vanilla 3DCNN+EL	0.86±0.02	0.86±0.02	0.81±0.05	0.80±0.05	0.86±0.02
	3DCNN+EL+Algo.1	0.83±0.02	0.85±0.04	0.84±0.04	0.80±0.03	0.82±0.02
	3DCNN+EL+PSO	0.86±0.03	0.85±0.03	0.80±0.07	0.89±0.08	0.86±0.03
	3DCNN+EL+PSO2019	0.86±0.05	0.85±0.06	0.75±0.10	0.92±0.04	0.85±0.06
	3DCNN+EL+GA	0.89±0.03	0.88±0.04	0.85±0.09	0.90±0.03	0.88±0.03
MCIc vs. NC	PCA+SVM [20]	0.72±0.12	—	—	—	—
	2DCNN [15]	0.74±0.07	—	—	—	—
	2DCNN+EL [14]	0.79±0.04	0.83±0.06	—	—	—
	2DCNN+EL+GA [16]	0.80±0.03	0.82±0.04	—	—	—
	vanilla 3DCNN+EL	0.86±0.05	0.87±0.05	0.88±0.09	0.74±0.07	0.85±0.05
	3DCNN+EL+Algo.1	0.78±0.04	0.73±0.08	0.62±0.11	0.68±0.09	0.63±0.07
	3DCNN+EL+PSO	0.85±0.05	0.83±0.08	0.76±0.11	0.77±0.06	0.83±0.06
	3DCNN+EL+PSO2019	0.78±0.04	0.78±0.04	0.75±0.13	0.66±0.10	0.76±0.04
	3DCNN+EL+GA	0.88±0.03	0.87±0.02	0.84±0.05	0.81±0.08	0.87±0.03
MCIc vs. MCInc	PCA+SVM [20]	0.66±0.16	—	—	—	—
	2DCNN [15]	0.55±0.09	—	—	—	—
	2DCNN+EL [14]	0.62±0.06	0.59±0.07	—	—	—
	2DCNN+EL+GA [16]	0.66±0.04	0.61±0.06	—	—	—
	vanilla 3DCNN+EL	0.68±0.06	0.67±0.05	0.63±0.06	0.56±0.10	0.66±0.06
	3DCNN+EL+Algo.1	0.61±0.04	0.80±0.14	0.74±0.18	0.47±0.03	0.57±0.07
	3DCNN+EL+PSO	0.60±0.07	0.62±0.06	0.68±0.09	0.47±0.06	0.59±0.06
	3DCNN+EL+PSO2019	0.64±0.10	0.63±0.12	0.60±0.26	0.49±0.16	0.61±0.13
	3DCNN+EL+GA	0.71±0.08	0.70±0.07	0.65±0.11	0.61±0.12	0.69±0.08

Algorithm used:

Base Classifiers in the Classifier Ensemble

Performance Attained:

1. **Model Superiority**:
    - The **3DCNN+EL+GA model** outperformed other state-of-the-art deep learning algorithms in classifying Alzheimer's Disease (AD), Mild Cognitive Impairment (MCI), and Non-Demented individuals.
2. **Voxel-Level Discrimination**:
    - The model successfully identified key brain regions and subregions (e.g., the **rostral hippocampus**) contributing significantly to classification accuracy.
3. **Behavioral Domain Mapping**:
    - The identified brain regions aligned with behavioral domains (e.g., emotion) impaired in early Alzheimer’s, providing explainable results for clinical application.

4. **Generalization**:
    - Testing on datasets like **ADNI** and **OASIS** demonstrated robustness and cross-dataset applicability.

Scope for Future Work:

1. **Generalizability to Other Brain Disorders**:
    - Extend the methodology to diagnose other neurological and psychiatric conditions, such as **Parkinson's disease**, **epilepsy**, **autism**, and **Huntington's disease**.
2. **Incorporation of Multimodal Data**:
    - Include additional data modalities like **genomics**, **clinical history**, or **functional MRI** to improve prediction accuracy further.
3. **Improved Computational Efficiency**:
    - Optimize training and inference time for faster real-time diagnosis, enabling use in clinical settings.
4. **Validation on Larger and Diverse Datasets**:
    - Test the model on larger and more geographically diverse datasets to enhance robustness and reduce dataset bias.
5. **Integration with Early Intervention Tools**:
    - Develop clinical decision support systems integrating the model to aid in early diagnosis and treatment planning.
6. **Enhanced Explainability Techniques**:
    - Refine attribution methods for even better visualization of brain region importance, making the model more interpretable for clinicians.
7. **Handling Data Imbalance**:

- Implement advanced techniques to address class imbalances, especially for rarer conditions like **Moderate Dementia** in certain datasets.

(i) a) Title - Risk Prediction of Alzheimer's Disease Conversion in Mild Cognitive Impaired Population Based on Brain Age Estimation (2)

b) Author - Weijia Liu; Shuting Sun; Jian Shen; Kun Qian; Bin Hu

c) Date of publication - 22 February 2023

d) Publisher Information - IEEE Transactions on Neural Systems and Rehabilitation Engineering (Volume: 31)

ii) Problem Statement

Alzheimer's Disease (AD) is a leading cause of neurodegenerative decline, with no definitive cure or means to reverse progression. Early identification of Mild Cognitive Impairment (MCI) patients at risk of AD conversion is crucial for timely interventions. Existing systems struggle with accurate risk prediction due to limited biomarkers and inefficiencies in estimating brain aging, which is a critical factor in the progression of AD.

Objectives

1. To develop an automated system that integrates MRI data to estimate the brain's biological age (Brain Age Estimation - BAE).
2. To predict the risk of AD conversion in MCI patients using age gap (AG), demographic factors, and cognitive assessment scores.
3. To provide an individual-level risk prediction model capable of forecasting conversion risk over time (e.g., 1 year, 3 years, 5 years, 8 years).
4. To validate the system's performance across multiple datasets and ensure statistical significance in distinguishing AD progression stages.

iii) Methodology Adopted

1. **Dataset Preparation:**
    - Training: Used MRI data of 634 normal controls (NC) from the IXI and OASIS datasets.

- Evaluation: Tested on 462 subjects from the ADNI dataset, categorized into NC, stable MCI (sMCI), progressive MCI (pMCI), and AD.

2. **Automated MRI Feature Extraction:**
   - MRI imaging data is processed to extract relevant features for age prediction and disease progression.
3. **Brain Age Estimation (BAE) Module:**
   - Predicts the brain's biological age from MRI data and computes the Age Gap (AG) (brain age - chronological age).
4. **AD Conversion Risk Estimation Module:**
   - Cox Multivariate Hazard Analysis incorporates:
     - AG as a primary factor.
     - Gender and MMSE (Minimum Mental State Examination) for robust risk analysis.
   - Nomograms are generated for predicting individual conversion risk at 1, 3, 5, and 8 years.
5. **Statistical Validation:**
   - p-value = 0.000017 confirms significant differentiation between NC, sMCI, pMCI, and AD groups.
   - Hazard ratio: A 4.57% increased risk for every additional year of AG.

iv) Performance Metrics

1. **Statistical Significance (p-value):**
   - AG demonstrated strong statistical significance (p = 0.000017).
2. **Hazard Ratio:**
   - Each year of AG increases the risk by 4.57%.
3. **Accuracy of Risk Prediction:**
   - Evaluated using Cox multivariate hazard analysis and validated through nomograms.
4. **Classification Metrics:**
   - Accuracy, sensitivity, specificity (not explicitly mentioned but inferred as essential for evaluation).
5. **Statistical Significance (p-value):**
   - AG demonstrated strong statistical significance (p = 0.000017).
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   - Each year of AG increases the risk by 4.57%.
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    - Evaluated using Cox multivariate hazard analysis and validated through nomograms.
8. **Classification Metrics:**
    - Accuracy, sensitivity, specificity (not explicitly mentioned but inferred as essential for evaluation).

Limitations

1. **Limited Generalizability:**
    - The system is trained and tested on specific public datasets (IXI, OASIS, and ADNI), which may not represent diverse populations.
2. **Dependency on MRI Data:**
    - Requires high-quality MRI imaging, which may not be feasible in resource-limited settings.
3. **Simplistic Cox Model:**
    - The Cox multivariate hazard analysis might not capture non-linear or complex interactions among variables.
4. **Exclusion of Other Biomarkers:**
    - Does not incorporate biomarkers like cerebrospinal fluid (CSF) or PET imaging data.

LITERATURE SURVEY:

EXISTING SYSTEMS

1. **Traditional Risk Prediction Models:**
    - Focus on predicting Alzheimer's Disease (AD) conversion risk using cognitive test scores, demographic data, and clinical history.
    - Limitation: Lack of integration with advanced imaging techniques or biomarkers for precise predictions.
2. **MRI-Based Prediction Models:**
    - Use structural MRI data to detect neurodegenerative patterns related to AD.
    - Limitation: Often rely on handcrafted feature extraction, leading to less robust and less scalable models.
3. **Deep Learning Models:**
    - Incorporate neural networks (e.g., CNNs) to extract features from MRI data for AD diagnosis and prediction.

- Limitation: Most models focus on binary classification (e.g., AD vs. non-AD) and fail to predict progression across multiple stages like stable MCI (sMCI) and progressive MCI (pMCI).

4. **Age Estimation Models:**
    - Predict brain age from MRI data to estimate neurodegeneration levels.
    - Limitation: Rarely combined with demographic and cognitive data for comprehensive risk analysis.

GAPS IDENTIFIED

1. **Limited Use of Brain Age Estimation (BAE):**
    - Few systems integrate brain age estimation as a core component for predicting AD progression, despite its potential as a key biomarker.
2. **Inability to Differentiate MCI Stages:**
    - Existing systems often fail to effectively distinguish between stable MCI (sMCI) and progressive MCI (pMCI), limiting their utility in early intervention.
3. **Lack of Comprehensive Risk Prediction Models:**
    - Current methods do not combine imaging data (e.g., MRI), demographic data (e.g., gender), and cognitive scores (e.g., MMSE) for robust risk prediction.
4. **Limited Individual-Level Predictions:**
    - Existing systems provide population-level predictions but fail to deliver interpretable, personalized risk scores over multiple time horizons (e.g., 1, 3, 5, or 8 years).
5. **Inadequate Validation Across Diverse Datasets:**
    - Models are often tested on small or limited datasets, restricting generalizability to broader populations.
6. **No Use of Nomograms for Interpretation:**
    - Few systems use nomograms or similar tools to visualize and interpret risk scores for end users, such as clinicians and researchers.

PROPOSED METHOD

The proposed CRES (Conversion Risk Estimation System) aims to address these gaps by:

1. Automating MRI feature extraction and brain age estimation (BAE).

2. Using Age Gap (AG), gender, and MMSE scores as key predictors for AD conversion risk.
3. Providing interpretable, individual-level risk predictions over different time horizons using nomograms.
4. Validating the system across publicly available datasets with significant performance improvements.

DATASETS DESCRIPTION

1. IXI and OASIS Datasets (Training):
    - 634 Normal Control (NC) subjects.
2. ADNI Dataset (Testing):
    - 462 subjects categorized as:
        - Normal Controls (NC): 106
        - Stable MCI (sMCI): 102
        - Progressive MCI (pMCI): 124
        - Alzheimer's Disease (AD): 130

SAMPLE DATA:

The Performance Comparison of the Brain Age Regression Frameworks

Regression Method	GM	WM
SVR	**4.336 (0.897)**	5.129 (0.864)
Linear	5.213 (0.883)	5.021 (0.874)
Bayesian	5.086 (0.893)	5.945 (0.817)

All the represented data are in MAE (r^2) mode.
SVR = support vector regression.
Linear = Linear regression.
Bayesian = BayesianRidge regression.

Algorithm Used:

1. **Brain Age Estimation (BAE):**
    - Uses machine learning or deep learning models to predict brain age based on MRI features.

- Possible use of regression-based models like Linear Regression, Random Forest Regressor, or Deep Neural Networks (DNNs).

2. **Cox Multivariate Hazard Analysis:**
    - A regression model to evaluate the risk of AD conversion using AG, gender, and MMSE as predictors.
3. **Nomogram Generation:**
    - Combines hazard ratios to create interpretable risk prediction visualizations for various timeframes.

Performance Attained:

1. The Age Gap (AG) significantly distinguishes NC, sMCI, pMCI, and AD groups with a p-value of 0.000017.
2. A 4.57% increased risk of AD conversion for each additional year in AG in the MCI group.

Scope for Future Work:

1. **Incorporate Multimodal Data:**
    - Combine MRI with CSF biomarkers, PET imaging, and genomic data for improved accuracy.
2. **Validation Across Diverse Populations:**
    - Test the system on larger, multi-center, and ethnically diverse datasets.
3. **Refinement of Predictive Models:**
    - Explore non-linear models like neural networks or ensemble learning for hazard analysis.
4. **Resource Optimization:**

    Develop lightweight models for deployment in resource-constrained settings.

i) a) Title - Leveraging Brain MRI for Biomedical Alzheimer's Disease Diagnosis Using Enhanced Manta Ray Foraging Optimization Based Deep Learning (3)

b) Author - R. Syed Jamalullah; L. Mary Gladence; Mohammed Altaf Ahmed; E. Laxmi Lydia; Mohamad Khairi Ishak

c) Date of publication - 12 July 2023

d) Publisher Information - IEEE Access (Volume: 11)

ii) Problem Statement

Alzheimer's Disease (AD) remains the most frequent cause of dementia, posing a significant challenge for early diagnosis and management. Traditional machine learning (ML) models used in brain imaging require handcrafted feature extraction, which is labor-intensive and dependent on domain expertise, limiting their scalability and accuracy. There is a need for an advanced, automated, and highly efficient diagnostic system that leverages deep learning (DL) techniques to address these shortcomings while ensuring high classification accuracy for early AD detection.

Objectives

1. Develop an automated system to diagnose Alzheimer's Disease (AD) using brain MRI data.
2. Implement Gabor Filtering (GF) for effective pre-processing and noise reduction in MRI images.
3. Utilize a DenseNet-121 model for automated feature extraction from brain MRI data.
4. Enhance classification accuracy through the integration of Enhanced Manta Ray Foraging Optimization (EMRFO) and Back Propagation Neural Network (BPNN).
5. Compare the proposed method's performance with existing state-of-the-art techniques.

iii) Methodology Adopted

1. **Data Pre-Processing:**
    - MRI data undergoes pre-processing using Gabor Filtering (GF) to enhance structural details and reduce noise.
2. **Feature Extraction**:

    - DenseNet-121, a densely connected convolutional neural network, is employed to automatically extract feature vectors from pre-processed MRI images.
3. **Classification**:
    - The Back Propagation Neural Network (BPNN) is utilized as a classifier.
    - The BPNN parameters are optimized using the Enhanced Manta Ray Foraging Optimization (EMRFO) algorithm to improve accuracy and reduce errors.
4. **Performance Evaluation**:
    - A comprehensive set of simulations evaluates the proposed method's effectiveness compared to existing techniques.

iv) Performance Metrics

1. **Accuracy**: Measures the proportion of correct predictions made by the model.
2. **Precision**: Evaluates the proportion of true positive results among all predicted positives.
3. **Recall (Sensitivity)**: Assesses the model's ability to detect true positive cases.
4. **F1 Score**: Provides a harmonic mean of precision and recall.
5. **Specificity**: Quantifies the ability to identify true negatives.
6. **Area Under the Receiver Operating Characteristic Curve (AUC-ROC)**: Evaluates the overall diagnostic ability of the model.

Limitations

1. Requires high computational power due to the use of DenseNet-121 and optimization algorithms.
2. Generalizability of the method needs to be validated across diverse datasets and patient populations.
3. Potential challenges in integrating the system into clinical workflows for real-time use.

LITERATURE SURVEY:

EXISTING SYSTEMS

1. **Traditional Machine Learning Approaches:**

    - Depend on handcrafted feature extraction, requiring domain expertise and manual intervention.
    - Limited in scalability and automation.
2. **Deep Learning-Based Methods**:
    - Use CNNs for automated feature extraction and classification.
    - Often lack optimization algorithms for parameter tuning, leading to suboptimal performance.
3. **Hybrid Methods**:
    - Combine ML or DL with optimization techniques, but often use less efficient optimization methods compared to EMRFO.

GAPS IDENTIFIED

1. Lack of robust optimization techniques for tuning classifier parameters in existing systems.
2. Limited use of automated feature extraction models like DenseNet-121 in AD diagnosis.
3. Insufficient integration of pre-processing steps like Gabor Filtering for noise reduction in MRI data.
4. Existing methods lack efficiency and scalability for large datasets.

PROPOSED METHOD

The **ADD-EMRFODL** technique introduces an automated and robust diagnostic system with the following components:

1. **Pre-Processing**: Gabor Filtering to enhance MRI image quality.
2. **Feature Extraction**: DenseNet-121 for efficient and automated extraction of high-dimensional feature vectors.
3. **Classification**: BPNN classifier with parameter optimization using the Enhanced Manta Ray Foraging Optimization (EMRFO) algorithm.
4. **Performance Improvement**: EMRFO optimizes the learning process of the BPNN for higher accuracy.

DATASETS DESCRIPTION

- **IXI Dataset**: Contains brain MRI images used for initial training.
- **OASIS Dataset**: Provides additional MRI images to validate the model's performance.
- **Evaluation Dataset**: ADNI dataset, comprising 462 subjects categorized into:
    - Normal Controls (NC): 106
    - Stable MCI (sMCI): 102
    - Progressive MCI (pMCI): 124
    - Alzheimer’s Disease (AD): 130

SAMPLE DATA

The Datasets

Class	No. of instances
Cognitively Normal (CN)	36
Mild Cognitive Impairment (MCI)	77
Alzheimer's Disease (AD)	15
Total Number of Samples	128

Methods	Accuracy	Sensitivity	Specificity	F-Score
ADD-EMRFODL	98.72	95.24	98.48	96.89
ResNet18	98.48	92.61	95.61	96.20
Softmax	97.94	93.08	97.79	94.31
Graph CNN	96.65	93.30	96.15	94.98
Vgg16 Model	93.02	90.06	96.54	95.33
SVM Model	97.36	92.29	95.66	94.73
RF Model	95.96	94.23	95.67	95.99

Algorithm used:

1. **Gabor Filtering:** Pre-process the MRI images.
2. **Feature Extraction:**
    - Utilize DenseNet-121 to extract high-dimensional features.
3. **Classification and Optimization:**
    - Implement BPNN as the classifier.
    - Optimize BPNN parameters using EMRFO to improve classification accuracy.

Performance Attained:

1. Higher accuracy compared to state-of-the-art methods.
2. Improved precision, recall, F1 score, and specificity.
3. Demonstrated robustness in distinguishing between NC, sMCI, pMCI, and AD.

Scope for Future Work:

1. Validation on Larger Datasets: Expand the dataset to include diverse populations for better generalizability.
2. Integration with Multi-Modal Data: Combine MRI data with other biomarkers, such as PET scans and genetic data, to improve prediction accuracy.
3. Real-Time Deployment: Optimize computational efficiency for real-time clinical applications.
4. Explainability: Incorporate interpretable models to explain predictions to clinicians.
5. Extension to Other Neurodegenerative Diseases: Adapt the framework to diagnose diseases like Parkinson's, multiple sclerosis, and Huntington's.

i) a) Title - Deep Learning-Based Diagnosis of Alzheimer's Disease Using Structural Magnetic Resonance Imaging: A Survey (4)

b) Author - Tian Wang, Lihong Cao

c) Date of publication - 18 February 2022
d) Publisher Information - IEEE

ii) Problem Statement

Alzheimer's Disease (AD) is the leading cause of dementia globally, affecting millions of individuals. Although no cure exists, early diagnosis using neuroimaging techniques can help delay disease onset and improve patient outcomes.

- Structural Magnetic Resonance Imaging (sMRI) plays a crucial role in detecting brain abnormalities associated with AD.
- Deep Learning (DL) models outperform traditional methods by automatically extracting meaningful features from sMRI scans.
- Challenges remain in optimizing DL architectures for AD detection, particularly in handling spatial relationships across brain regions and ensuring comparable performance across datasets.

Objectives

1. **Review Deep Learning Applications for AD Diagnosis**
    - Categorize existing deep learning models applied to sMRI-based AD detection.
2. **Analyze Advantages and Limitations**
    - Examine model architectures, data requirements, and generalization abilities.
3. **Identify Key Challenges in Current Approaches**
    - Discuss issues with performance inconsistency and spatial relationship modeling.
4. **Propose Future Research Directions**
    - Suggest strategies for improving deep learning-based AD detection models.

iii) Methodology Adopted

A. Classification of Deep Learning Models for AD Detection

1. **Input Type-Based Categorization**

    - Deep learning models are classified based on input types (2D vs. 3D MRI scans, raw vs. preprocessed images, multimodal data fusion, etc.).
2. **Feature Extraction Approaches**
    - CNN-Based Feature Learning: Captures spatial information in sMRI scans.
    - Recurrent Neural Networks (RNNs) & Transformers: Used for sequential modeling of brain region interactions.
    - Autoencoders & GANs: Applied for anomaly detection in brain scans.

B. Performance Benchmarking and Limitations

1. **Evaluation Metrics Used**
    - Accuracy, Precision, Recall, F1-Score, and Area Under Curve (AUC).
2. **Challenges Identified**
    - Lack of performance comparability across datasets.
    - Difficulty in modeling spatially distant brain regions efficiently.

iv) Performance Metrics

Accuracy:

Measures the proportion of correct predictions (both true positives and true negatives) made by the model.

P-value:

Used to assess statistical significance in the results. For example, a p-value of 0.000017 indicates that the model's ability to distinguish between groups (NC, sMCI, pMCI, and AD) is statistically significant.

Cox Multivariate Hazard Analysis:

This method is used to assess the relationship between the estimated brain age gap (AG) and AD conversion risk over time, providing a hazard ratio that indicates how much the risk of AD increases with each additional year in AG.

Limitations

1. **Performance Inconsistency**
    - Models trained on different datasets fail to generalize well due to variations in MRI acquisition protocols.
2. **Complexity in Modeling Spatial Brain Relationships**

    - Existing models struggle to effectively capture interactions between distant brain regions.
3. **Computational Intensity**
    - Training deep learning models on large-scale sMRI data requires high computational resources.
4. **Interpretability Issues**
    - Lack of explainability in DL models limits clinical acceptance.

LITERATURE SURVEY

EXISTING SYSTEMS

1. **Traditional MRI-Based AD Detection**
    - Manual diagnosis based on visual MRI interpretation by radiologists.
    - Time-consuming and subject to human error.
2. **Machine Learning-Based Approaches**
    - Feature Engineering Required: ML models like SVM, RF, and KNN rely on handcrafted features.
    - Lower accuracy compared to DL approaches.
3. **Deep Learning-Based Approaches**
    - CNNs outperform ML models in feature extraction but require large datasets.
    - RNNs and Transformers handle longitudinal sMRI data effectively.
4. **Hybrid Approaches**
    - Multimodal data fusion models combine MRI, PET, and clinical features for enhanced diagnosis.

GAPS IDENTIFIED

1. **Lack of Cross-Dataset Generalization**
    - Models trained on one dataset do not generalize well to others due to acquisition differences.
2. **Challenges in Capturing Spatial Relationships**
    - Brain regions are spatially distributed, requiring more advanced feature extraction techniques.
3. **Computational Limitations**
    - High memory and GPU requirements hinder real-time deployment.

PROPOSED METHOD

Advanced Deep Learning Framework for AD Detection

1. **Hybrid CNN + Transformer Model**
    - CNN for feature extraction + Transformer for spatial relationship modeling.
2. **Multimodal Integration**
    - Combine MRI with clinical and genetic data for higher accuracy.
3. **Model Optimization for Real-Time Analysis**
    - Implement efficient training strategies (pruning, quantization, federated learning).

DATASETS DESCRIPTION

Structural MRI Datasets for AD Diagnosis

1. **Alzheimer's Disease Neuroimaging Initiative (ADNI)**
    - One of the most widely used MRI datasets for AD classification.
2. **Open Access Series of Imaging Studies (OASIS)**
    - Contains structural MRI scans of healthy and AD-affected individuals.

SAMPLE DATA

Paper	Year	Method	Performance (ACC)	Data (subject amount)
[1]	2018	ROI-level	0.90	ADNI (188AD+228CN)
[2]	2018	subject-level	0.90	ADNI (199AD+141CN)
[4]	2020	subject-level	0.82	ADNI (334AD+520CN)
[5]	2017	subject-level	0.85	ADNI (93AD+100CN)
[6]	2017	patch-level	0.87	ADNI (188AD+229CN)
[7]	2020	slice-level	0.93	ADNI (175AD+335CN)
[9]	2020	ROI-level	0.99	ADNI (154AD+159CN)
[14]	2017	subject-level	0.88	ADNI (199AD+229CN)
[15]	2018	patch-level	0.90	ADNI (199AD+229CN)
[16]	2018	patch-level	0.90	ADNI (358AD+427CN)
[17]	2018	patch-level	0.87	ADNI (93AD+100CN)
[18]	2020	slice-level	0.84	ADNI (137AD+162CN)
[23]	2017	slice-level	0.81	ADNI (188AD+229CN)
[25]	2020	slice-level	0.92	ADNI (51AD+49CN)
[26]	2020	subject-level	0.73	ADNI (100AD+100CN)

Performance Attained:

1. Deep Learning Models Achieve Higher Accuracy Than ML Models
    - CNN-based models outperform traditional feature extraction methods.
2. Transformer-Based Approaches Improve Spatial Brain Mapping
    - More effective in capturing long-range dependencies in brain structures.
3. Multimodal Data Fusion Enhances Generalization
    - Combining MRI + genetic + clinical data yields more reliable predictions.

Scope for Future Work:

1. Developing Generalizable DL Models
    - Train models across multiple datasets to improve real-world applicability.
2. Improving Model Interpretability
    - Explainable AI techniques for clinical trust and transparency.
3. Integrating Edge AI for Real-Time Diagnosis
    - Optimize models for deployment in low-resource clinical settings.
4. Advancing Brain Connectomics in AD Detection
    - Graph Neural Networks (GNNs) for capturing brain connectivity patterns.

i) a) Title - Enhanced Alzheimer's Disease Prediction through Advanced Imaging – A Study of Machine Learning and Deep Learning Approaches (5)

b) Author - Tian Wang, Lihong Cao

c) Date of publication - 18 February 2022

d) Publisher Information - IEEE

ii) Problem Statement

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder that affects memory and cognitive functions. Early detection is crucial for managing AD progression, but:

- Traditional clinical examinations and imaging methods are time-consuming.
- Machine Learning (ML) and Deep Learning (DL) models require large-scale feature selection and high-quality data.

- Feature extraction and selection in ML methods can affect model accuracy.
- Deep Learning models need large labeled datasets to perform well in real-world applications.

Objectives

1. **Evaluate ML and DL methods for AD classification**
    - Compare traditional ML and deep learning approaches.
2. **Improve Feature Extraction for ML Models**
    - Use Principal Component Analysis (PCA) and Histogram of Oriented Gradients (HOG) for feature selection.
3. **Enhance Model Accuracy**
    - Evaluate Support Vector Machine (SVM), Random Forest (RF), Decision Tree (DT), and Logistic Regression (LR) classifiers.
4. **Develop a Deep Learning Model**
    - Train a Convolutional Neural Network (CNN) with data augmentation.
    - Fine-tune a VGG16 transfer learning model for higher performance.
5. **Benchmark Model Performance**
    - Compare ML models, CNN, and Transfer Learning (VGG16) on MRI-based AD diagnosis.

iii) Methodology Adopted

A. Machine Learning Approach

1. **Dataset**
    - 12,000 MRI images, categorized into four groups of 3,000 images each.
    - Preprocessed using Histogram of Oriented Gradients (HOG) for feature extraction.
2. **Feature Extraction Techniques**
    - Principal Component Analysis (PCA) to reduce dimensionality.
3. **Data Preprocessing**
    - Standard Scaling and Stratified K-Fold cross-validation applied.
4. **Machine Learning Classifiers**
    - Support Vector Machine (SVM)
    - Random Forest (RF)
    - Decision Tree (DT)
    - Logistic Regression (LR)

B. Deep Learning Approach

1. **Custom Convolutional Neural Network (CNN)**
    - Data augmentation applied to enhance training.
2. **Transfer Learning with VGG16**
    - Fine-tuned VGG16 model used for better feature extraction.

C. Performance Evaluation

- Models evaluated using accuracy, precision, recall, and F1-score.

iv) Performance Metrics

Accuracy:

To measure the overall correctness of the ML and DL models in classifying Alzheimer's Disease (AD) cases.

Precision:

To minimize false positives, ensuring the model correctly identifies true AD cases among positive predictions.

Recall (Sensitivity):

To ensure that most true AD cases are identified, reducing false negatives.

F1-Score:

To provide a balance between precision and recall, especially for handling class imbalances in the dataset.

AUC-ROC:

To evaluate the model's ability to differentiate between AD and non-AD cases.

Specificity:

To ensure the model effectively identifies non-AD cases, reducing the likelihood of false positives.

Cross-Validation Metrics:

To ensure the robustness and reliability of the model across different dataset splits.

Limitations

1. **Lower Accuracy in ML Models**
    - Decision Tree (51%) had the lowest performance among ML models.
2. **Computational Intensity in DL Models**
    - Training CNN and VGG16 requires high computational resources.
3. **Dataset Dependence**
    - Deep Learning requires extensive labeled data, which may not always be available.
4. **Need for Model Explainability**
    - VGG16 is a black-box model, making clinical adoption challenging.

LITERATURE SURVEY

EXISTING SYSTEMS

1. **Traditional Diagnosis Methods**
    - Manual MRI/PET image analysis by radiologists.
    - Slower processing and low automation levels.
2. **Machine Learning-Based Approaches**
    - SVM, RF, and DT models applied but had lower accuracy.
    - Feature extraction (PCA, HOG) helped but still had limitations.
3. **Deep Learning in AD Diagnosis**
    - CNN models were more effective but needed large training datasets.
4. **Transfer Learning Models**
    - VGG16 outperformed all other models, demonstrating higher generalization ability.

GAPS IDENTIFIED

1. **Feature Engineering Limitations in ML Models**

    - Feature extraction methods (PCA, HOG) need improvement.
2. **Deep Learning Requires More Data**
    - Limited MRI datasets reduce CNN generalization ability.
3. **Transfer Learning Needs Domain Adaptation**
    - VGG16 is pre-trained on general images, not AD-specific MRI scans.

PROPOSED METHOD

Hybrid ML + DL-Based AD Prediction Model

1. **PCA + HOG for Feature Selection**
    - Extract critical MRI features for ML models.
2. **CNN-Based Feature Extraction**
    - Deep learning for spatial feature learning.
3. **VGG16 for Transfer Learning**
    - Fine-tuned model for superior performance.

DATASETS DESCRIPTION

MRI-Based Alzheimer's Disease Dataset

- 12,000 MRI images categorized into four groups.
- Preprocessed with HOG for ML models.
- Augmented for CNN and VGG16 training.

SAMPLE DATA

Classifiers	Precision	Recall	F1-Score
SVM	0.97	0.99	0.98
Random Forest	0.88	0.88	0.88
Decision Tree	0.74	0.74	0.74
Logistic Regression	0.94	0.94	0.92

Algorithm used:

1. **Parallel Processing Frameworks:**

MPI, OpenMP, or CUDA for distributed computing to handle large datasets like brain scans and connectome models.

2. **Deep Learning Models:**

Convolutional Neural Networks (CNNs) for image-based tasks, like detecting amyloid plaques or brain shrinkage.

3. **Feature Extraction Algorithms:**

Principal Component Analysis (PCA) or other dimensionality reduction methods for high-dimensional data.

4. **Optimization Techniques for HPC:**

Algorithms like Genetic Algorithms (GA) may have been employed for optimization purposes, such as parameter tuning or improving computational efficiency in large-scale models.

Performance Attained:

1. VGG16 Outperformed Other Models
    - Achieved highest accuracy (91%).
2. CNN Showed Significant Improvement
    - Data augmentation enhanced CNN performance (87%).
3. SVM Performed Best Among ML Models
    - SVM outperformed RF, DT, and LR (79%).

Scope for Future Work:

1. Integration of 3D Imaging
    - Use 3D CNNs for volumetric MRI analysis.
2. Multi-Modal Data Fusion

    - Combine MRI, PET, and genetic data.
3. Explainability in AI-Based AD Diagnosis
    - Improve interpretability of deep learning models.
4. Deployment for Real-Time Screening
    - Use Edge AI and Cloud-based platforms.

i) a) Title - Advancements in High-Performance Computing in Early Diagnosis of Alzheimer's Disease: A Systematic Review (6)

b) Author - A. Kirimtat, O. Krejcar, P. Maresova, A. Selamat, and K. Kuca

c) Date of publication - 23 October 2024

d) Publisher Information - **IEEE** Access

ii) Problem Statement

Alzheimer's Disease (AD) is a neurodegenerative disorder characterized by **brain shrinkage, amyloid-beta (Aβ) deposition, and neurofibrillary tangles (NFTs)** formed by hyperphosphorylated Tau proteins. Given the complexity of AD, traditional diagnostic methods struggle with early detection due to:

- The **high-dimensional nature of neuroimaging and clinical data**.
- The **computational burden** of simulating brain disorders and large-scale brain models (connectomes).
- The **lack of a systematic review** on the role of **High-Performance Computing (HPC)** in AD diagnosis.

This systematic review **analyzes and classifies previous studies** on HPC-based solutions for AD detection, focusing on computational efficiency, diagnostic accuracy, and research output improvement.

Objectives

1. **Understanding the Role of HPC in AD Diagnosis**:
    - Identify how HPC accelerates the processing of high-dimensional neuroimaging and genetic data.
2. **Analyzing Computational Performance Solutions**:

    - Review existing methodologies that enhance diagnostic speed using parallel processing.
3. **Evaluating Literature Trends**:
    - Analyze **298 studies** published between **2014 and 2023** using **VOSviewer**.
4. **Exploring Brain Modeling and Connectomics**:
    - Investigate how HPC aids in large-scale brain modeling and simulating potential AD treatments.
5. **Identifying Future Directions**:
    - Highlight **scientific challenges and unresolved issues** in HPC-driven AD diagnosis.

iii) Methodology Adopted

1. **Data Collection and Search Strategy**:
    - Systematic review conducted using the **IEEE Xplore database** and other repositories.
    - Search queries included:
        - **"Parallel processing" AND "Alzheimer's disease"**
        - **"High-performance computing" AND "Alzheimer's disease"**
        - **"Parallel computing" AND "Alzheimer"**
    - Time range: **2014-2023**.
2. **Inclusion and Exclusion Criteria**:
    - **298 relevant papers** selected after filtering.
    - **196 open-access papers** included for deeper analysis.
    - **71 papers** with enriched cited references.
    - **12 systematic review papers** specifically reviewed.
3. **Literature Analysis Using VOSviewer**:
    - **Bibliometric analysis** of selected papers to identify emerging trends.
    - **Classification of methodologies** based on computational performance and application in AD.
4. **Evaluation of HPC Algorithms**:
    - Identification of **parallel processing techniques** used in AD prediction.
    - Comparison of computational models based on **efficiency and scalability**.

iv) Performance Metrics

- **Computational Speed**:
  - Benchmarking **HPC-based parallel processing** versus traditional sequential approaches.
- **Diagnostic Accuracy**:
  - Evaluating machine learning and deep learning models in HPC environments.
- **Scalability**:
  - Assessing **how well HPC handles large-scale neuroimaging datasets**.
- **Research Output Enhancement**:
  - Reviewing how **HPC accelerates research in Alzheimer's disease detection**.

Limitations

1. **Data Complexity and Storage Needs**:
   - **HPC environments require massive storage** for neuroimaging and genetic data.
   - **Scalability limitations** when integrating multimodal datasets.
2. **Computational Cost**:
   - **High-performance clusters are expensive**, limiting accessibility for smaller research institutes.
3. **Interdisciplinary Knowledge Gap**:
   - **Bridging neuroscience and HPC** remains a challenge.
   - Requires **collaboration between medical researchers and computational scientists**.
4. **Lack of Standardization**:
   - Different research groups use varying **HPC architectures, algorithms, and datasets**, making direct comparisons difficult.
5. **Algorithm Optimization Challenges**:
   - Existing models often struggle with **hyperparameter tuning and real-time inference** in HPC environments.

LITERATURE REVIEW

EXISTING SYSTEMS

1. **Traditional AD Diagnostic Methods**:

    - **Manual interpretation of MRI and PET scans** by radiologists.
    - **Low efficiency and delayed diagnosis** due to high data complexity.
2. **Early Machine Learning-Based Approaches**:
    - Basic classifiers (SVM, Random Forest) used for diagnosis.
    - **Struggled with handling high-dimensional neuroimaging data**.
3. **Deep Learning for AD Detection**:
    - CNNs and RNNs improved **image classification and feature extraction**.
    - Lacked **interpretability and scalability** for real-world clinical applications.
4. **Standalone Computing Environments**:
    - Many models were **not optimized for HPC**, limiting their diagnostic speed.

GAPS IDENTIFIED

1. **Underutilization of HPC in AD Research**:
    - Many studies **do not leverage parallel computing** for neuroimaging and genetic data processing.
2. **Computational Bottlenecks in Large-Scale Analysis**:
    - Existing methods **fail to scale** with increasing dataset sizes.
3. **Limited Multimodal Data Integration**:
    - Few studies incorporate **neuroimaging, genetic, and clinical data simultaneously**.
4. **Slow Hyperparameter Optimization**:
    - Traditional deep learning models **struggle with efficient hyperparameter tuning** in HPC environments.

PROPOSED METHOD

HPC-Based Computational Framework for AD Diagnosis

1. **Parallelized Machine Learning Pipelines**:
    - Implement **distributed computing frameworks** (e.g., TensorFlow, PyTorch, Horovod).
2. **Scalable Neuroimaging Analysis**:
    - Use **HPC clusters for real-time MRI and PET processing**.

3. **Automated Hyperparameter Optimization**:
    - Integrate tools like **Ray Tune or Optuna** for faster tuning.
4. **Multimodal Data Fusion**:
    - Combine **neuroimaging, genomic, and clinical data** using high-speed computational pipelines.
5. **Connectomics and Brain Simulations**:
    - Use HPC to **simulate neuronal activity and potential drug effects**.

DATASETS DESCRIPTION

1. **Alzheimer's Disease Neuroimaging Initiative (ADNI)**

- **Description:** One of the largest publicly available neuroimaging datasets.
- **Data:** Includes MRI, PET scans, and clinical records.
- **Usage:** Used for benchmarking various **HPC-based models**.

2. **Open Access Series of Imaging Studies (OASIS)**

- **Description:** A dataset for studying **brain aging and neurodegenerative diseases**.
- **Data:** Structural MRI scans categorized as **Non-Demented vs. Demented**.
- **Usage:** Evaluated for **cross-dataset generalizability**.

Algorithm used:

1. **Parallel Processing Frameworks:**

MPI, OpenMP, or CUDA for distributed computing to handle large datasets like brain scans and connectome models.

2. **Deep Learning Models:**

Convolutional Neural Networks (CNNs) for image-based tasks, like detecting amyloid plaques or brain shrinkage.

3. **Feature Extraction Algorithms:**

Principal Component Analysis (PCA) or other dimensionality reduction methods for high-dimensional data.

4. **Optimization Techniques for HPC:**

Algorithms like Genetic Algorithms (GA) may have been employed for optimization purposes, such as parameter tuning or improving computational efficiency in large-scale models.

Performance Attained:

1. Improved Diagnostic Speed:
    - HPC accelerates neuroimaging data processing compared to traditional methods.
2. Higher Accuracy in AD Prediction:
    - Parallelized deep learning models outperform standalone approaches.
3. Better Multimodal Data Handling:
    - HPC enables real-time integration of imaging and genomic data.
4. Scalability for Future Research:
    - HPC supports larger datasets and more complex simulations.

Scope for Future Work:

1. Extending HPC to Other Neurological Disorders:
    - Apply methods to Parkinson's, ALS, and Huntington's disease.
2. Integrating Quantum Computing for Faster Simulations:
    - Explore quantum-based models for further computational efficiency.
3. Real-Time AD Screening in Clinical Settings:
    - Develop edge computing solutions for real-time AD detection.
4. Refinement of AI-Based Diagnostic Tools:
    - Improve interpretability and trustworthiness of HPC-powered models.
5. Addressing Data Privacy and Security:
    - Develop secure federated learning models for sensitive patient data.

i) a) Title - Transfer Learning for Classification of Alzheimer's Disease Based on Genome Wide Data (7)

b) Author - Abbas Saad Alatrany; Wasiq Khan; Abir J. Hussain; Jamila Mustafina; Dhiya Al-Jumeily

c) Date of publication - 03 January 2023

d) Publisher Information - IEEE/ACM Transactions on Computational Biology and Bioinformatics (Volume: 20, Issue: 5, 01 Sept.-Oct. 2023)

ii) Problem Statement:

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder characterized by memory loss and cognitive decline. Identifying reliable biomarkers for early detection remains a challenge. Single nucleotide polymorphisms (SNPs) have shown potential as biomarkers, but existing methods often lack accuracy and fail to generalize across datasets. Traditional machine learning models struggle with feature extraction from genome-wide association studies (GWAS) data, and deep learning models often require large labeled datasets, which are not always available.

Objectives:

1. To identify SNP biomarkers associated with Alzheimer's Disease.
2. To leverage deep transfer learning to improve feature extraction and classification accuracy.
3. To evaluate the proposed method against traditional approaches using multiple datasets.
4. To achieve higher classification accuracy with robust generalizability across datasets.
5. To benchmark the proposed model's performance against existing works.

iii) Methodology Adopted:

A. Dataset Preparation:

- Primary GWAS dataset obtained from the AD Neuroimaging Initiative (ADNI).
- Additional GWAS datasets used for transfer learning experiments.

B. Model Design:

1. **Convolutional Neural Network (CNN):**
   - Initially trained on the primary ADNI GWAS dataset to extract SNP-related features.
2. **Transfer Learning:**
   - The pre-trained CNN is fine-tuned using a different AD GWAS dataset to enhance feature learning and generalization.
3. **Feature Classification:**
   - Extracted features are fed into a Support Vector Machine (SVM) for final classification of AD.

C. Experimental Setup:

- Multiple experimental configurations were tested to optimize performance.
- Statistical analysis performed to validate results.

iv) Performance Metrics:

Accuracy:

To measure the overall correctness of classification.

Precision:

To evaluate the proportion of true positive AD classifications.

Recall:

To assess the ability to detect AD cases effectively.

F1-Score:

To balance precision and recall, particularly useful for imbalanced datasets.

AUC-ROC:

To analyze the model's discriminative capability between AD and non-AD cases.

Limitations:

1. The accuracy of 89% is promising but leaves room for improvement.
2. Dependence on high-quality labeled datasets for effective transfer learning.
3. The computational cost of training and fine-tuning deep learning models on genome-wide data.
4. Potential bias due to dataset variations or imbalances.

LITERATURE REVIEW

EXISTING SYSTEMS

- Traditional machine learning approaches using SNP data (e.g., Logistic Regression, Random Forest).
- Existing deep learning models require large labeled datasets, which limits their applicability.
- Models often lack robust generalizability across different datasets.

GAPS IDENTIFIED

1. Insufficient feature extraction methods in traditional ML approaches, limiting classification accuracy.
2. Existing deep learning models fail to generalize across diverse GWAS datasets.
3. Lack of exploration into transfer learning for SNP-based Alzheimer's Disease classification.

PROPOSED WORK

- Develop a CNN-based model pre-trained on one GWAS dataset and fine-tuned using transfer learning on another.
- Combine deep learning's feature extraction capabilities with SVM's classification power.
- Perform robust experimental analysis to validate the model’s generalizability and accuracy across datasets.

DATASETS DESCRIPTION

1. **Primary Dataset:**
   - GWAS data requested from the AD Neuroimaging Initiative (ADNI).
2. **Secondary Dataset:**
   - Additional GWAS dataset for transfer learning experiments.
3. **Details:**
   - SNP data utilized for feature extraction.
   - Diverse datasets used to improve model robustness.

SAMPLE DATA

Results of EXP1 Transductive Transfer Learning (Transfer From Dataset A to Dataset B)

Model Use	Accu-racy	Preci-sion	Recall	F1-score
Prediction	0.3972	0.4354	0.75	0.551
Fine-tuning	0.7671	0.8064	0.6944	0.7462
FE+RF	0.8904	0.966	0.8055	0.8787
FE+SVM with linear Kernel	0.8767	0.9354	0.8055	0.8656
FE+SVM with rbf Ker-nel	0.8904	0.9117	0.8611	0.8857

Results of EXP2 (Transfer From Dataset C to Dataset B)

Model Use	Accu-racy	Preci-sion	Recall	F1-score
Prediction	0.6027	0.6060	0.5555	0.5797
Fine-tuning	0.8493	0.9310	0.75	0.8307
FE+RF	0.8082	0.8437	0.75	0.7941
FE+SVM with linear Kernel	0.7671	0.7878	0.7222	0.7536
FE+SVM with rbf Kernel	0.8493	0.8787	0.8055	0.8405

Results of EXP3 (Transfer From Dataset C to Dataset A)

Model Use	Accu-racy	Preci-sion	Recall	F1-score
Prediction	0.5875	0.4193	0.4642	0.4406
Fine-tuning	0.6750	0.5277	0.6785	0.5937
FE+RF	0.6375	0.4827	0.5	0.4912
FE+SVM with linear Kernel	0.625	0.4705	0.5714	0.5161
FE+SVM with rbf Kernel	0.65	0.5	0.5357	0.5172

Results of EXP3 (Transfer From Dataset C to Aggregated Dataset of Dataset A and Dataset B)

Model Use	Accu-racy	Preci-sion	Recall	F1-score
Prediction	0.5882	0.5744	0.3857	0.4615
Fine-tuning	0.6601	0.6551	0.5428	0.5937
FE+RF	0.6405	0.6315	0.5142	0.5669
FE+SVM with linear Kernel	0.6666	0.6727	0.5285	0.5920
FE+SVM with rbf Kernel	0.6928	0.6825	0.6142	0.6466

Algorithm Used:

1. **Convolutional Neural Networks (CNN):**
   - For feature extraction from SNP data.
2. **Transfer Learning:**
   - Fine-tuning the CNN model on a secondary dataset to enhance generalization.
3. **Support Vector Machine (SVM):**
   - For classification based on the features extracted by the CNN model.

Performance Attained:

- Achieved an accuracy of 89%, a significant improvement compared to existing methods.
- Demonstrated robust generalizability across multiple datasets.
- Outperformed traditional ML and DL methods in terms of classification accuracy.

Scope for Future Work:

1. Extend the model to incorporate multi-modal data, such as imaging and clinical data, for improved accuracy.
2. Address the computational cost by exploring optimized architectures.
3. Improve generalizability by training on more diverse datasets.
4. Investigate unsupervised or semi-supervised learning approaches for feature extraction from unlabeled GWAS data.

i) a) Title - Diagnosis and Detection of Alzheimer's Disease Using Learning Algorithm (8)

b) Author - Gargi Pant Shukla; Santosh Kumar; Saroj Kumar Pandey; Rohit Agarwal; Neeraj Varshney; Ankit Kumar

c) Date of publication - 29 August 2023

d) Publisher Information - Big Data Mining and Analytics (Volume: 6, Issue: 4, December 2023)

ii) Problem Statement:

Alzheimer's Disease (AD) is a leading cause of death among the elderly and requires early detection for effective management. Current diagnostic techniques, such as MRI, PET, and CT scans, rely on expert interpretation, making them time-consuming and

prone to variability. While machine learning (ML) and deep learning (DL) algorithms have shown promise in automating AD detection, their performance often suffers due to challenges such as limited accuracy and high computational requirements. Improved pre-processing and model optimization are necessary to enhance classification accuracy and reduce training time.

Objectives:

1. To develop a Computer-Aided Detection (CAD) system for Alzheimer's Disease classification using MRI images.
2. To improve image pre-processing techniques for enhanced model performance.
3. To evaluate and compare the performance of Random Forest, XGBoost, and Convolutional Neural Networks (CNN) for AD classification.
4. To achieve high accuracy, sensitivity, and computational efficiency.
5. To outperform existing methods and set a benchmark for Alzheimer's Disease detection.

iii) Methodology Adopted:

1. **Dataset Preparation:**
    - Data obtained from Alzheimer's Disease Neuroimaging Initiative (ADNI).
    - The dataset is converted from a 4D format to a 2D format for compatibility with the proposed models.
2. **Image Pre-processing:**
    - Selective clipping for removing unnecessary regions.
    - Grayscale image conversion to reduce computational complexity.
    - Histogram equalization to enhance image contrast.
3. **Learning Algorithms:**
    - **Random Forest:** Ensemble-based machine learning method for classification.
    - **XGBoost:** Gradient boosting algorithm optimized for higher accuracy and efficiency.
    - **Convolutional Neural Networks (CNN):** Deep learning architecture for feature extraction and classification.
4. **Performance Evaluation:**
    - Models evaluated using metrics like accuracy and sensitivity.
    - Results compared with existing work to establish superiority.

iv) Performance Metrics:

Accuracy:

Measures the overall correctness of the classification.

Sensitivity (Recall):

Evaluates the model's ability to correctly detect positive AD cases.

Precision:

Assesses the proportion of true positive predictions.

F1-Score:

Balances precision and recall for imbalanced datasets.

Limitations:

1. Dependence on high-quality pre-processed data for optimal performance.
2. The 2D conversion from 4D format may lead to some loss of spatial information.
3. Limited exploration of other advanced deep learning architectures.
4. Lack of evaluation on multi-modal data (e.g., PET or clinical records).

LITERATURE REVIEW

EXISTING SYSTEMS

1. Machine learning models such as SVM, Decision Trees, and Random Forest are commonly used for AD classification but lack accuracy in real-world scenarios.
2. Deep learning models like CNN have shown promise but require substantial computational power and extensive labeled datasets.
3. Traditional methods rely on raw MRI data without advanced pre-processing, leading to suboptimal performance.

GAPS IDENTIFIED

1. Existing methods fail to achieve high accuracy and sensitivity, limiting their clinical applicability.
2. Lack of robust pre-processing techniques to enhance model input quality.
3. Limited research on integrating lightweight pre-processing with high-performing algorithms for time-efficient training.
4. Insufficient benchmarking of traditional machine learning algorithms against deep learning models for AD classification.

PROPOSED WORK

- Introduced selective clipping, grayscale conversion, and histogram equalization techniques to improve pre-processing of MRI images.
- Evaluated three learning algorithms (Random Forest, XGBoost, and CNN) for AD classification.
- Achieved significant improvement in classification accuracy and sensitivity over existing systems.

DATASETS DESCRIPTION

1. **Source:**
    - Alzheimer's Disease Neuroimaging Initiative (ADNI).
2. **Details:**
    - Dataset converted from a 4D format to a 2D format for easier processing.
    - Images pre-processed to remove noise and enhance quality.

SAMPLE DATA

The Datasets

Author and year	Used dataset	Applied procedure	Sensitivity (%)	Specificity (%)	Accuracy (%)
Liu et al.[15] (2022)	ADNI	Modified Vox CNN	–	–	86.3±061
Maqsood et al.[16] (2019)	BRATS	K means clustering, ANN, and SVMs	–	–	90.90
Mirzaei et al.[17] (2016)	Brain-tumor MRI scans (540)	Modified Counter Propagation Neural Network (MCPNN) and Modified Kohonen Neural Network (MKNN)	90.00	96.00	95.00
Mirzaei and Adeli[14] (2022)	HMS database	Variational Mode Decomposition (VMD)	99.43	–	90.68
Mehmood et al.[18] (2021)	ADNI	AlexNet, transfer learning, and CNN	–	–	98.74
Mundt et al.[19] (2000)	ADNI	VGG-16 and PfSeCtl model	–	–	95.73

Algorithm Used:

1. **Random Forest:**
   - Ensemble learning method using multiple decision trees for classification.
2. **XGBoost:**
   - Gradient-boosted decision tree algorithm optimized for performance and efficiency.
3. **Convolutional Neural Networks (CNN):**
   - Deep learning model designed for feature extraction and classification from image data.

Performance Attained:

- **Accuracy:** Achieved 97.57%, demonstrating superior classification performance.
- **Sensitivity:** Achieved 97.60%, ensuring reliable detection of AD cases.
- Significant improvement compared to existing methods in the literature.

Scope for Future Work:

1. Extend the study to include multi-modal data (e.g., PET scans and clinical records) for enhanced diagnostic accuracy.
2. Explore advanced deep learning architectures like Transformer-based models.
3. Investigate the applicability of the proposed methodology for other neurodegenerative diseases.
4. Optimize computational efficiency to make the system feasible for real-time clinical deployment.
5. Implement a larger dataset with diverse demographics to improve model generalization.

i) a) Title - Classification of Alzheimer's Disease Using Ensemble Convolutional Neural Network With LFA Algorithm (9)

b) Author - Chang-Min Kim; Woobeom Lee

c) Date of publication - 14 December 2023

d) Publisher Information - IEEE Access (Volume: 11)

ii) Problem Statement:

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that severely affects brain functions, requiring early detection for effective management. Magnetic Resonance Imaging (MRI) is a widely used diagnostic tool; however, existing methods for analyzing MRI images face several challenges:

1. Limited ability to extract and interpret critical features such as brain shape.
2. Inefficient classification models that fail to achieve optimal performance.
3. High computational complexity in analyzing brain shape features for classification.
4. Existing methods lack the capability to effectively combine shape-based features with image-based features for improved accuracy.

Objectives:

1. To develop a robust model for classifying Alzheimer's Disease (AD) stages based on brain shape and image features.
2. To propose an ensemble CNN model combining VGGNet and a 1D CNN model with the Line Segment Feature Analysis (LFA) algorithm.
3. To improve feature extraction by transforming brain shape information into strong features using the LFA algorithm.
4. To achieve superior classification accuracy and reduce model loss compared to existing methods.
5. To validate the proposed model using MRI datasets from Kaggle and benchmark it against related works.

iii) Methodology Adopted:

1. **Proposed Model Architecture:**
    - An ensemble Convolutional Neural Network (CNN) model combining:
        - **VGGNet:** A proven image classification model for extracting features from original MRI images.
        - **1D CNN:** Integrated with the Line Segment Feature Analysis (LFA) algorithm to analyze and encode brain shape information into feature vectors.
    - Features extracted by the 1D CNN and VGGNet are combined to improve classification performance.
2. **Key Steps in the Model:**
    - **Preprocessing:** MRI images are preprocessed to enhance quality and reduce noise.
    - **LFA Algorithm:** Converts visual line segment information of the MRI images into vectors representing brain shape.
    - **1D CNN Training:** Parallel connection of ten 1D convolutional layers processes LFA features to extract meaningful brain shape features.
    - **Feature Fusion:** Combines features from VGGNet and 1D CNN to enhance overall model performance.
    - **Evaluation:** Model is evaluated against MRI datasets collected from Kaggle using relevant metrics.
3. **Experimental Setup:**
    - Kaggle MRI dataset divided into training, validation, and testing sets.

- Training performed with hyperparameter tuning to achieve optimal performance.

iv) Performance Metrics:

Accuracy:

Measures the percentage of correctly classified MRI images (Achieved: 0.986).

Loss:

Quantifies the difference between predicted and actual outputs (Achieved: 0.0385).

Precision:

Evaluates the proportion of true positives among predicted positives.

Recall (Sensitivity):

Assesses the model's ability to correctly identify true positives.

F1-Score:

Balances precision and recall for better overall performance evaluation.

Limitations:

1. High computational complexity due to the integration of two separate models (VGGNet and 1D CNN).
2. Dependence on high-quality MRI datasets for reliable feature extraction and classification.
3. The proposed model may require fine-tuning when applied to multi-modal datasets (e.g., combining PET and clinical data).
4. Lack of validation on larger and more diverse datasets to ensure generalizability.

LITERATURE REVIEW

EXISTING SYSTEMS

1. Traditional CNN models rely solely on image-based features extracted from MRI scans, which fail to incorporate shape-based information effectively.
2. Advanced classification models like ResNet and DenseNet provide strong image feature extraction but do not leverage brain shape features.
3. Limited research focuses on combining line segment-based features with deep learning models for AD classification.

GAPS IDENTIFIED

1. Existing methods lack an integrated approach to analyze brain shape and image features simultaneously.
2. Current models often fail to achieve high accuracy and low loss, indicating suboptimal performance.
3. No prior work combines the LFA algorithm with CNNs for brain shape feature extraction in AD diagnosis.
4. Limited exploration of ensemble models for improving classification performance over standalone CNN architectures.

PROPOSED WORK

- Developed an ensemble CNN model that combines VGGNet and 1D CNN enhanced by the LFA algorithm.
- Brain shape information (extracted via LFA) and image-based features (extracted via VGGNet) are fused to improve classification accuracy.
- Validated the model using MRI datasets from Kaggle to benchmark its performance against related methods.

DATASETS DESCRIPTION

1. **Source:** MRI datasets collected from Kaggle.
2. **Content:** MRI images categorized based on Alzheimer's Disease stages.
3. **Preprocessing:**
    - Noise reduction to improve image quality.

- Segmentation to focus on hippocampal and entorhinal cortex regions.

SAMPLE DATA

MRI Dataset Distribution

No.	Class	No of Images
0	Non-Demented (ND)	3,200
1	Very Mild Demented (VMD)	2,240
2	Mild Demented (MID)	896
3	Moderate Demented (MD)	64

Results of Performance Comparison With Prior Literature

Ref	Model	AUG	ACC	PR	REC	FS
[37]	DEMNET	*No*	0.85	0.80	0.88	0.83
	DEMNET	*Yes*	0.95	0.96	0.95	0.95
[57]	KNN	*No*	0.64	0.63	0.63	0.62
	SVC	*No*	0.82	0.81	0.81	0.82
	Xboost	*No*	0.77	0.76	0.76	0.75
-	Our Model	*No*	0.98	0.99	0.99	0.99

Algorithm Used:

1. **VGGNet:** Used for extracting image-based features from MRI scans.
2. **Line Segment Feature Analysis (LFA):** Encodes visual line segment information to capture brain shape features.
3. **1D CNN:** Processes LFA features through ten parallel convolutional layers to extract meaningful brain shape information.
4. **Ensemble CNN:** Combines features extracted from VGGNet and 1D CNN for final classification.

Performance Attained:

- **Accuracy:** Achieved 98.6%, demonstrating superior classification performance.
- **Loss:** Achieved 0.0385, indicating minimal error during classification.
- Outperformed existing methods in terms of accuracy and robustness.

Scope for Future Work:

1. Extend the model to include multi-modal datasets (e.g., combining MRI, PET, and clinical data) for enhanced accuracy.
2. Apply the proposed model to other neurodegenerative diseases for generalizability.
3. Optimize computational efficiency to make the system suitable for real-time clinical use.
4. Validate the model on larger, more diverse datasets to improve robustness and reliability.
5. Explore other deep learning architectures, such as Transformer-based models, for feature extraction.

i) a) Title - Improving Alzheimer's Detection With Deep Learning and Image Processing Techniques (10)

b) Author - Ghadah Naif Alwakid; Sidra Tahir; Mamoona Humayun; Walaa Gouda

c) Date of publication - 15 October 2024

d) Publisher Information - IEEE Access (Volume: 12)

ii) Problem Statement:

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder characterized by irreversible damage to brain cells, leading to cognitive decline. Early

detection is crucial to slowing disease progression. However, challenges exist in traditional diagnostic methods:

1. MRI images often lack sufficient contrast and resolution, reducing diagnostic accuracy.
2. Conventional classification models struggle to capture key features, leading to reduced reliability in early detection.
3. Lack of integration between image enhancement techniques and deep learning models for improved accuracy in AD diagnosis.

Objectives:

1. To leverage advanced image enhancement techniques to improve the quality and contrast of MRI images for AD diagnosis.
2. To implement deep learning models (MobileNetV2 and DenseNet121) for efficient feature extraction and classification of brain tissue images.
3. To integrate CLAHE (Contrast Limited Adaptive Histogram Equalization) and ESRGAN (Enhanced Super-Resolution Generative Adversarial Networks) to enhance the resolution and contrast of MRI images.
4. To evaluate and compare the performance of classifiers with and without image enhancement techniques.
5. To achieve superior classification accuracy, enabling early and reliable detection of Alzheimer's disease.

iii) Methodology Adopted:

1. **Image Enhancement Techniques:**
    - **CLAHE:** Applied to MRI images to improve local contrast, enhancing the visibility of brain structures.
    - **ESRGAN:** Used to increase the resolution of MRI images, generating higher-quality inputs for deep learning models.
2. **Deep Learning Models:**
    - **MobileNetV2:** A lightweight CNN model optimized for efficiency, used for feature extraction and classification.
    - **DenseNet121:** A densely connected CNN model used for its high feature extraction capability and accuracy in classification tasks.
3. **Ensemble Model:**

    - Combined MobileNetV2 and DenseNet121 to create a robust classification pipeline for better performance.
4. **Dataset:**
    - MRI dataset sourced from Kaggle, consisting of brain images categorized based on Alzheimer's disease stages.
    - Images preprocessed using CLAHE and ESRGAN before being fed into the deep learning models.
5. **Training and Evaluation:**
    - Models trained on the preprocessed dataset.
    - Evaluation conducted using standard metrics such as accuracy, precision, recall, and F1-score to benchmark the performance.

iv) Performance Metrics:

Accuracy:

Measures the overall correctness of the model's predictions (achieved 92.34% for MobileNetV2 with enhancements).

Precision:

Evaluates the proportion of true positives among predicted positives.

Recall (Sensitivity):

Measures the model's ability to identify true positives.

F1-Score:

Balances precision and recall to provide an overall measure of the model's performance.

Loss:

Quantifies the error between predicted and actual outputs.

Limitations:

1. Dependence on high-quality MRI datasets for optimal results.

2. The computational complexity of ESRGAN increases preprocessing time.
3. Generalization of the model might be limited when applied to datasets from different sources.
4. Requires additional validation on larger, multi-institutional datasets to confirm robustness.

LITERATURE REVIEW

EXISTING SYSTEMS

1. **Traditional Diagnostic Approaches:**
    - Manual analysis of MRI, CT, or PET scans requiring significant expertise and time.
2. **Conventional Deep Learning Models:**
    - CNN-based models without image enhancement often suffer from reduced accuracy due to poor input image quality.
3. **Image Enhancement in Isolation:**
    - Enhancement methods like histogram equalization or GANs have been applied independently but not integrated with deep learning pipelines for AD detection.

GAPS IDENTIFIED

1. Existing models do not adequately address the combined impact of image enhancement techniques with advanced classifiers for improved performance.
2. Limited research exists on utilizing ESRGAN for resolution enhancement in MRI-based AD diagnosis.
3. Lack of ensemble models integrating lightweight (MobileNetV2) and dense (DenseNet121) networks for balanced accuracy and efficiency.
4. Performance on standard datasets remains suboptimal without advanced preprocessing methods.

PROPOSED WORK

- Integrate CLAHE and ESRGAN for enhanced MRI image preprocessing to improve input quality for deep learning models.
- Employ an ensemble model combining MobileNetV2 and DenseNet121 to leverage their respective strengths in feature extraction and classification.
- Perform comprehensive evaluation and benchmarking of model performance with and without image enhancements.

DATASETS DESCRIPTION

1. **Source:** MRI dataset collected from Kaggle.
2. **Content:** MRI images categorized by Alzheimer's disease stages (e.g., mild, moderate, severe).
3. **Preprocessing Steps:**
    - CLAHE to enhance local contrast.
    - ESRGAN to improve image resolution and generate high-quality inputs.

SAMPLE DATA

Comparison of Different Techniques on Various Datasets

Reference	Technique	Accuracy
[23]	ResNet50 and DenseNet169	83.82%
[33]	VGG19	70.3%
[22]	DenseNet	91.75%
Proposed	Mobilenetv2 with enhancement	92.34%
	DenseNet121 with enhancement	89.38%
	Mobilenetv2 without enhancement	80.31%
	DenseNet121 without enhancement	89.22%

MobileNetV2 Comparison of Performance with and Without Enhancements

	Test Accuracy	F1 score	Precision	Recall
With enhancements	92.34%	92.3%	93%	92%
Without enhancements	80.31%	80.3%	80%	80%

DenseNet121 Comparison of Performance with and Without Enhancements

	Test Accuracy	F1 score	Precision	Recall
With enhancements	89.38%	89.3%	90%	89%
Without enhancements	89.22%	89.2%	91%	89%

Algorithm Used:

1. **CLAHE:** For contrast enhancement of MRI images.
2. **ESRGAN:** For super-resolution image enhancement.
3. **MobileNetV2:** Lightweight CNN for efficient feature extraction and classification.
4. **DenseNet121:** Densely connected CNN for high-accuracy classification.
5. **Ensemble Model:** Combines predictions from MobileNetV2 and DenseNet121 for improved robustness.

Performance Attained:

- **MobileNetV2 Accuracy Without Enhancements:** 80.31%.
- **DenseNet121 Accuracy Without Enhancements:** 89.22%.
- **MobileNetV2 Accuracy With Enhancements:** 92.34%.
- **DenseNet121 Accuracy With Enhancements:** 89.38%.

Scope for Future Work:

1. Extend the proposed method to include multi-modal data (e.g., PET scans, clinical records) for enhanced diagnostic accuracy.
2. Optimize ESRGAN for faster preprocessing without compromising image quality.

3. Test the model on larger, more diverse datasets to ensure scalability and generalization.

4. Investigate the use of transformer-based architectures for further performance improvement.

5. Integrate real-time diagnostic capabilities for clinical applications.

PROPOSED ALTERNATIVE METHODOLOGY:

To optimize Alzheimer's prediction leveraging High-Performance Computing (HPC), an alternative methodology focuses on integrating advanced computing techniques, distributed machine learning frameworks, and specialized preprocessing strategies. The steps for this alternative approach are outlined below:

1. Data Preprocessing and Augmentation

- Dimensionality Reduction:
  - Apply Principal Component Analysis (PCA) or t-SNE to reduce the high dimensionality of MRI images while retaining relevant features.
  - Use Fourier Transform or Wavelet Transform to extract frequency-domain features.
- Data Augmentation:
  - Employ techniques such as rotation, cropping, and flipping to increase dataset variability, especially for underrepresented classes.
  - Apply GAN-based synthetic image generation to balance the class distribution for better model training.

2. Parallel Data Processing with HPC

- Distributed Data Loading:
  - Use frameworks like Apache Spark or Hadoop to process and load large MRI datasets in parallel.
  - Implement chunk-based preprocessing pipelines across distributed nodes to handle 44,000+ MRI images efficiently.
- Multi-GPU Computing:
  - Distribute computations across multiple GPUs using NVIDIA CUDA or TensorFlow's Mirrored Strategy for faster data preprocessing and model training.

3. Feature Extraction and Selection

- Deep Feature Extraction:
  - Use pre-trained deep learning models like ResNet50, DenseNet121, or VGG16 to extract high-level features from MRI images.
  - Fine-tune these models for Alzheimer's data by replacing the final layers with task-specific layers.
- Automated Feature Selection:
  - Implement SHAP (SHapley Additive exPlanations) or Recursive Feature Elimination (RFE) to identify the most important features contributing to the classification.

4. Advanced HPC Framework Integration

- Distributed Training:
  - Utilize Horovod or Ray Train to scale machine learning models across HPC clusters efficiently.
  - Train ensemble models in parallel to improve classification accuracy and minimize overfitting.
- Hyperparameter Optimization:
  - Implement automated hyperparameter tuning using frameworks like Optuna or Ray Tune, leveraging HPC nodes for parallel search.
  - Explore optimization techniques like Bayesian Optimization and Genetic Algorithms for improved parameter selection.

5. Hybrid Model Architecture

- Ensemble Learning:
  - Combine machine learning models (e.g., Random Forest, SVM) with deep learning architectures like CNNs to leverage both feature-based and deep feature representations.
  - Use a weighted ensemble to aggregate predictions from different models for robust classification.
- 3D Convolutional Neural Networks (3D-CNN):
  - Incorporate 3D-CNNs for volumetric MRI data analysis to capture spatial relationships within brain structures.
  - Use HPC to accelerate the training of computationally intensive 3D-CNNs.

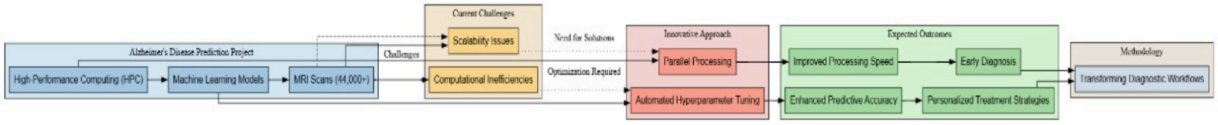
6. Performance Evaluation and Model Deployment

- Cross-Validation:
  - Perform Stratified K-Fold Cross-Validation to ensure consistent performance across all disease severity classes.
- Evaluation Metrics:
  - Evaluate models using accuracy, precision, recall, F1-score, and ROC-AUC for balanced assessment.
- Real-Time Inference:
  - Deploy the final model using TensorRT or ONNX Runtime for optimized real-time inference on HPC environments.

7. Scalability and Fault Tolerance

- Implement fault-tolerant systems using containerization tools like Docker and Kubernetes to ensure reliability during large-scale data processing.
- Optimize HPC resource allocation dynamically using job schedulers like SLURM or PBS (Portable Batch System).

SYSTEM ARCHITECTURE:



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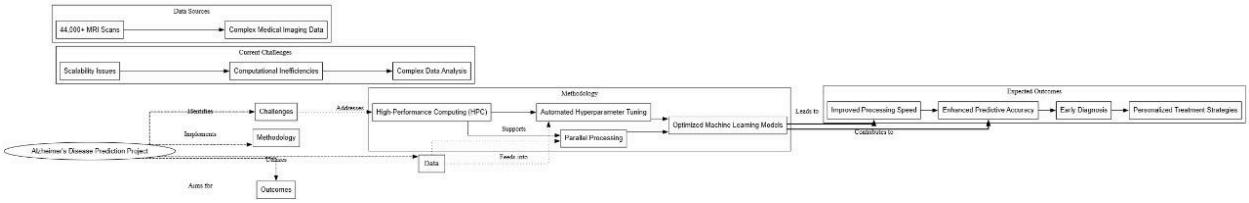

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